Toward Robust Detector-Level Inference for JWST/NIRISS SOSS Transmission Spectroscopy

Davide Staub

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Abstract

Transmission spectroscopy uses wavelength-dependent transit depths to probe the opacity and composition of exoplanet atmospheres. Transmission spectra are inferred through a sequence of detector calibration, background treatment, spectral extraction, light-curve construction, and transit fitting. Independent pipelines can produce different spectra from the same observation, while agreement cannot establish accuracy because the true spectrum of a real planet is unknown. This project asks how accurately and robustly JWST/NIRISS SOSS transmission spectra can be recovered from detector-level time series, and whether joint detector-level inference can reduce biases associated with background treatment and extraction-first analysis. Two connected tools are being developed. The first is a detector-level injector that produces realistic SOSS observations with a known transmission spectrum and delivers identical data to independent recovery methods. The second is NOVA, a forward model that fits the time-dependent stellar transit, overlapping spectral orders, instrument response, and additive background in a common detector likelihood. Current development has established a reproducible injector and end-to-end diagnostic comparison, and has identified instrument-response support and source-background allocation as the main limitations. In one opened synthetic case, NOVA recovered the common $R = 100$ spectrum more accurately than the Ahsoka comparison, but NOVA did not satisfy the strict convergence criteria and the result is not science ready. The remaining PhD programme will complete the response-consistent benchmark, expand the test matrix across atmospheric and detector conditions, apply the framework to additional SOSS observations, extend it to emission and phase curves, and combine selected visits and JWST instruments in joint atmospheric inference. The planned outputs are a benchmark and methods paper, target-specific studies where spectral changes alter atmospheric conclusions, a cross-instrument synthesis, and the PhD thesis.

1 Introduction and Literature Review

The transmission spectrum of an exoplanet is never measured directly. It is inferred from detector-level time series through a chain of calibration, background treatment, spectral extraction, light-curve construction, and transit fitting, and every atmospheric conclusion drawn from it inherits the quality of that inference. Independent analyses of the same observation can produce different spectra, and their agreement cannot establish accuracy, because a real planet never provides an independently known spectrum against which the recovered one can be scored. This project responds with two connected tools: a detector-level injector that generates realistic JWST/NIRISS SOSS observations with a known transmission spectrum, and NOVA, a forward model that infers the wavelength-dependent transit jointly with instrument response and additive background in a single detector likelihood.

This introduction develops that argument in five steps. Section 1.1 reviews why exoplanet atmospheres are primary scientific targets and how transmission spectroscopy observes them. Section 1.2 introduces JWST, the NIRISS/SOSS mode, and the WASP-17 b observation whose three independent published reductions form the initial case study. Section 1.3 follows the conventional extraction-first workflow from detector reads to a transmission spectrum and shows why neither agreement nor disagreement between such analyses can measure recovery accuracy. Section 1.4 identifies the two structural choices, background treatment and early spectral compression, that motivate joint detector-level inference and the benchmark. Section 1.5 states the research question and objectives.

1.1 Exoplanet atmospheres and transmission spectroscopy

The discovery of planets beyond the Solar System has shown that planetary systems can look very different from our own. These exoplanets span a wide range of sizes, temperatures, and orbital configurations, including many with no close Solar System counterpart [Winn and Fabrycky, 2015]. Understanding this diversity requires determining the physical and chemical properties of individual planets and assessing how reliably those properties can be inferred from observations.

For orientation, exoplanets are often described using four broad observational categories: terrestrial planets, super-Earths, Neptune-like planets, and gas giants [NASA Science, 2026]. These categories provide useful descriptive shorthand, but their boundaries are not fundamental, and a category does not uniquely determine a planet's interior or atmospheric composition.

For well-characterised transiting planets, mass and radius are usually among the first physical properties inferred. Together they constrain bulk density, but planets with similar masses and radii can have different interior structures and atmospheric mass fractions, leaving their exact compositions underconstrained [Rogers and Seager, 2010]. Distinguishing between these possibilities requires observables that respond directly to the material and energy in a planet's outer layers. Atmospheres provide those observables [Madhusudhan, 2019].

Atmospheric composition and structure reflect present conditions and may retain information about a planet's formation and evolution. Atomic and molecular abundances constrain chemical inventories, elemental ratios, and atmospheric metallicity. These quantities may carry information about the gas and solids accreted during formation, although they do not determine a unique formation history [Öberg et al., 2011]. Temperature structures, clouds, and hazes govern how radiation is absorbed, emitted, and transported, while escape signatures trace atmospheric loss driven by the host star's high-energy irradiation [Vidal-Madjar et al., 2003]. When interpreted alongside mass, radius, age, and irradiation, atmospheric measurements can therefore test hypotheses about formation, migration, enrichment, and long-term mass loss [Madhusudhan, 2019]. Atmospheric characterisation provides a route from detecting a planet to testing physical explanations for its present state. To do so reliably, however, the atmospheric signal must first be recovered from the light of the host star.

The atmospheric signal discussed above can be observed when a planet passes in front of its host star and some of the starlight crosses the planet's atmospheric limb. This observing method is transmission spectroscopy. It is the starting point for the present work because its repeated, well-defined transit geometry provides a tractable setting in which to test how detector modelling and reduction choices affect a recovered atmospheric spectrum.

Transmission spectroscopy measures the wavelength dependence of a transit. Ignoring limb darkening for the moment, the transit depth is approximately

$$\delta(\lambda) \simeq \left[\frac{R_{\mathrm{p}}(\lambda)}{R_\star}\right]^2. \qquad (1)$$

The effective planetary radius $R_{\mathrm{p}}(\lambda)$ increases where the atmosphere is more opaque and decreases where it is more transparent. Absorption by atoms, molecules, clouds, and aerosols can therefore imprint wavelength-dependent structure on the measured transit depth [Brown, 2001, Seager and Sasselov, 2000]. Figure 1 summarises this geometry and its connection to a transmission spectrum.

Only the starlight passing through a thin annulus around the planet probes its atmosphere, so the wavelength-dependent atmospheric signal is a small fraction of the total stellar flux. The signal is stronger when the planet is large relative to its host star and when its atmosphere extends over a greater range of altitudes. High temperatures, atmospheres dominated by light gases such as hydrogen and helium, and weak gravity favour an extended atmosphere, whereas clouds and hazes can obscure spectral features. Both the atmosphere and the geometry of the planet-star system therefore determine how readily a transmission signal can be measured [Brown, 2001, Seager and Sasselov, 2000].

One class of planet with especially favourable properties for transmission spectroscopy is the hot Jupiter. Hot Jupiters are a subgroup of gas giants that orbit close to their host stars and are consequently strongly irradiated. Their large radii, high temperatures, hydrogen-dominated atmospheres, and sometimes low gravities make their transmission signatures comparatively strong [Brown, 2001, Madhusudhan, 2019]. The first detection of an exoplanet atmosphere identified sodium absorption in the hot Jupiter HD 209458 b [Charbonneau et al., 2002]. Hot Jupiters are therefore useful targets for developing and testing methods that recover transmission spectra from spectroscopic time series. A stronger signal makes errors in reduction and light-curve fitting easier to diagnose before moving to planets

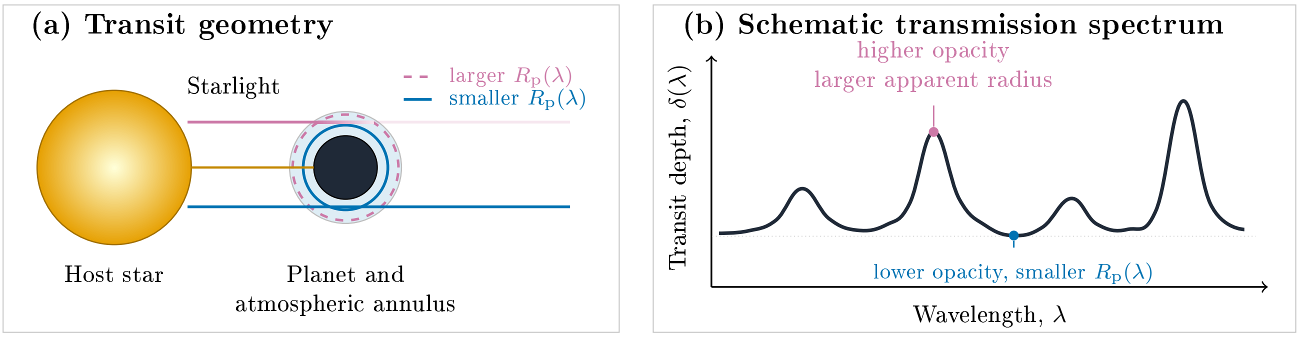

Figure 1: Principle of transmission spectroscopy. (a) During transit, the opaque planet blocks starlight while a small fraction passes through its atmospheric annulus. The altitude at which the atmosphere becomes opaque, and therefore the apparent planetary radius, depends on wavelength. (b) Greater atmospheric opacity produces a larger $R_p(\lambda)$ and a deeper transit, whereas lower opacity produces a shallower transit. The plotted spectrum is schematic and does not represent a particular planet or instrument.

with intrinsically weaker atmospheric signatures. Hot Jupiters are a starting point rather than an end point. The initial case study in this project is WASP-17 b, an unusually low-density hot Jupiter for which a complete JWST/NIRISS SOSS transit and several independent reductions are available [Anderson et al., 2010, Louie et al., 2025]. Once a method has been validated on this and other **such** favourable targets, it can be applied to smaller and cooler planets, including the worlds most relevant to questions of planetary habitability.

Transmission is one of three principal time-series geometries used for atmospheric spectroscopy. Secondary-eclipse **: secondary-eclipse** measurements isolate the planet's dayside emission by comparing the combined star and planet flux with the stellar flux measured while the planet is hidden [Coulombe et al., 2023, Deming et al., 2005, Gressier et al., 2024]. Phase **[Coulombe et al., 2023, Deming et al., 2005, Gressier et al., 2024], while phase** curves follow the changing planetary contribution over an orbit and constrain day-night temperature contrast, heat redistribution, and longitudinal structure [Kempton et al., 2023, Knutson et al., 2007, Mikal-Evans et al., 2023, Splinter et al., 2025]. Each geometry measures a different weighting of the atmosphere. This project begins with transmission spectroscopy. Emission **; emission** and phase-curve analyses are planned future applications of the same detector-level framework.

Across all three geometries, the atmosphere is not observed in isolation. Its signal is embedded in light from the host star and modified by the telescope, detector, observing geometry, and data analysis. Different combinations of abundance, temperature, reference radius, clouds, hazes, and stellar heterogeneity can also produce similar spectral signatures [Madhusudhan, 2019, Rackham et al., 2018]. Atmospheric characterisation is consequently an inverse problem: the data constrain a family of physical explanations rather than measuring a unique atmospheric state directly [Barstow and Heng, 2020]. Any atmospheric interpretation can therefore only be as reliable as the spectrum supplied to the retrieval.

The transmission spectrum is itself an inferred data product. Detector measurements are calibrated and modelled to estimate transit depth as a function of wavelength. A separate atmospheric forward model or retrieval then maps that spectrum to composition, temperature, clouds, and other physical quantities [Brown, 2001, Madhusudhan, 2019]. These two inference stages answer different questions. A retrieval can be statistically precise while inheriting a bias from the recovered spectrum, and a well-recovered spectrum can still admit several atmospheric interpretations. The present work concerns the first stage: how accurately the transmission spectrum is recovered.

1.2 JWST, NIRISS/SOSS, and the WASP-17 b case study

The first confirmed exoplanet system was found through pulsar timing. Arecibo observations revealed periodic timing variations from planets orbiting the millisecond pulsar PSR**targets for such measurements came from three decades of discovery. Pulsar timing revealed the first confirmed** exoplanets around PSR B1257+12 [Wolszczan and Frail, 1992]. Three years later, ELODIE , and the radial-velocity measurements established detection of 51Pegasi b as the first planet found around a solar-type star [Mayor and Queloz, 1995]. Its short orbital period and giant-planet mass revealed a Pegasi b established a class of close-in giant planet that was not anticipated from the architecture of the Solar System . These discoveries established that other planetary systems exist and can differ markedly from our own, but isolated detections could not show how common different systems are.

Kepler changed the scale of exoplanet science by [Mayor and Queloz, 1995]. Kepler transformed isolated detections into a population, monitoring approximately 150,000 stars for transits [Borucki et al., 2010]. Its discoveries enabled population and enabling measurements of planet occurrence, size, period, and system architecture [Winn and Fabrycky, 2015] . TESS retained the transit method but [Borucki et al., 2010, Winn and Fabrycky, 2015] , while TESS surveyed nearly the whole sky , prioritising bright and and prioritised the bright, nearby host stars that are most favourable for mass measurements and atmospheric follow-up [Ricker et al., 2015]. Alongside discoveries from other facilities, these surveys helped expand expanded the confirmed census beyond 6,000 exoplanets , with thousands of additional candidates awaiting validation [NASA Exoplanet Archive, 2026]. They established the population context for individual systems and supplied many of the targets for atmospheric study.

[NASA Exoplanet Archive, 2026]. The Hubble Space Telescope (HST) demonstrated that those and Spitzer demonstrated that such targets could be studied spectroscopically. Excess sodium absorption during the transit of : the sodium detection in HD 209458 b provided the first detection of an exoplanet atmosphere [Charbonneau et al., 2002]. Spitzer secondary-eclipse observations then detected thermal emission from was followed by Spitzer thermal-emission measurements of HD 209458 b and TrES-1 b [Charbonneau et al., 2005, Deming et al., 2005], while a Spitzer phase curve of HD 189733 b mapped the changing brightness of the planet's visible hemisphere over its orbit [Knutson et al., 2007]. Later HST and Spitzer programmes moved beyond individual detections. For example, Sing et al. [2016] combined 0.3–5 μm observations of ten hot Jupiters and [Knutson et al., 2007], and comparative programmes such as the ten-planet survey of Sing et al. [2016], which found a continuum from clear to cloudy atmospheres. These studies established the transmission, emission, and phase-curve techniques used today, but their broad wavelength coverage generally had to be assembled from restricted instrument bandpasses observed at different epochs.

JWST was designed as a 6.5-metre, cryogenic, infrared-optimised observatory with instruments spanning the near- to mid-infrared [Gardner et al., 2006]. Unlike Kepler and TESS, it is not a wide-field transit survey. For : for the time-series spectroscopy considered here, JWST it follows up planets that have already been discovered and whose transit times are already known, using its observing time to characterise their atmospheres. Its collecting areaincreases the number of detected photons, its cold opticsreduce thermal background over much of the infrared, and its instrument suite gives , cold optics, and instrument suite give access to multiple molecular bands over a much broader wavelength range than a single HST or Spitzer mode. Commissioning , and commissioning showed that the observatory's sensitivity, stability, and spectral performance met or exceeded the mission requirements in most areas [Rigby et al., 2023]. JWST can therefore measure smaller wavelength-dependent changes in transit depth while measuring covering several atmospheric absorbers and continuum regions within individual observing modes and visits. This reduces, although it does not eliminate, the need to combine measurements obtained with different instruments or at different epochs.

JWST carries four science instruments: NIRCam, NIRSpec, MIRI, and NIRISS [Albert et al., 2023, Gardner et al., 2006]. Several of these instruments support exoplanet time-series spectroscopy. This project begins with the NIRISS Single Object Slitless Spectroscopy (SOSS) modebecause the initial , and with a single observation: a complete transit of WASP-17 btransit has , an unusually low-density hot Jupiter, for which three independent published reductions that permit a controlled comparison [Louie et al., 2025]. Its [Anderson et al., 2010, Louie et al., 2025] . The mode's slitless, cross-dispersed design produces curved spectral orders that overlap on the detector, making SOSS a demanding test of spectral extraction [Albert et al., 2023].

The GR700XD grism produces cross-dispersed orders covering approximately 0.6–2.8 μm, with resolving power near $R = 650$ at 1.25 μm [Albert et al., 2023]. A weak cylindrical lens spreads the spectrum over roughly 23 detector rows in the cross-dispersion direction. Distributing the light over more pixels prevents any one pixel from saturating as quickly and therefore permits observations of bright host stars [Albert et al., 2023]. The bandpass contains several water bands, the potassium doublet, and regions that can constrain other absorbers and optical scattering by clouds and hazes, with the precise sensitivity depending on the target atmosphere [Albert et al., 2023, Holmberg and Madhusudhan, 2023, Louie et al., 2025].

Figure 2 shows the curved first- and second-order footprints in a median SOSS detector image.

JWST time-series spectroscopy must distinguish changes in transit depth at the level of tens to hundreds of parts per million. During commissioning, an in-flight observation of the standard star BD+60°1753 reached approximately 20 parts per million scatter after removal of a long-term trend and averaging over 40-minute intervals [Albert et al., 2023]. This on-sky test demonstrated the time-series stability of SOSS, but it was not a transit observation and did not measure the precision of wavelength-dependent transit depths.

The 20 parts per million value describes a white-light time series that combines photons across the full SOSS bandpass. A transmission spectrum instead measures the transit depth separately in narrower wavelength channels. Each channel therefore contains only part of the total photon count and normally has greater photon noise than the corresponding white-light measurement [Albert et al., 2023, Louie et al., 2025]. For WASP-17 b, the three reductions reported median uncertainties of 46–73 parts per million per spectral channel across the two useful orders [Louie et al., 2025]. This range does not identify which reduction is more accurate, but it demonstrates that the precision assigned to a transmission spectrum is analysis-dependent at a scale relevant to the atmospheric signal.

The first SOSS exoplanet analyses demonstrated that the mode can recover rich atmospheric structure. The WASP-39 b Early Release Science spectrum contained multiple water bands, potassium absorption, and evidence of clouds [Feinstein et al., 2023]. All JWST time-series spectroscopy reductions must address detector calibration, bad pixels and cosmic rays, background emission, readout structure, wavelength calibration, spectral assignment, limb darkening, and time-dependent noise. SOSS adds a distinctive combination of curved and physically overlapping orders, slitless contamination by field-star spectra, and a spatially structured zodiacal background [Bell et al., 2022, Holmberg and Madhusudhan, 2023, Radica et al., 2023]. The atmospheric information is scientifically valuable, but recovering it requires choices about detector calibration, extraction, and light-curve fitting. Strong signals provide leverage for diagnosing those choices, while the curved and overlapping traces expose assumptions in the detector model. This combination makes SOSS a useful test case for comparing reduction methods.

WASP-17 b is the initial development case rather than the complete scope of the project. Its low density and low gravity give it an extended atmosphere [Anderson et al., 2010]. Alderson et al. [2022] combined HST STIS and WFC3 spectra with Spitzer IRAC photometry to construct a 0.3–5 μm transmission spectrum, but the atmospheric interpretation retained distinct high- and low-metallicity solutions. Louie et al. [2025] then analysed a complete NIRISS/SOSS transit using three independent reductions: Ahsoka, which they introduced; `transitspectroscopy` [Espinoza, 2022] with `juliet` [Espinoza et al., 2019]; and supreme-SPOON [Radica et al., 2023], now distributed as exoTEDRF [Radica, 2024]. The three resulting spectra agreed closely over most of the bandpass, and the wider SOSS coverage constrained the water abundance more tightly than the earlier HST and Spitzer data allowed [Louie et al., 2025]. This combination of earlier observations, a complete SOSS transit, and multiple reductions of the same data makes WASP-17 b a strong case for controlled method comparison. Later benchmark stages will include other targets and injected atmospheric scenarios.

1.3 From detector integrations to the validation problem

JWST does not record a transmission spectrum directly. In a NIRISS/SOSS time-series observation, the detector is read nondestructively several times within each integration. These individual reads are called groups. Charge accumulates across the groups, and a ramp fit estimates a count rate for each pixel in each integration. Repeating this process throughout a visit produces a

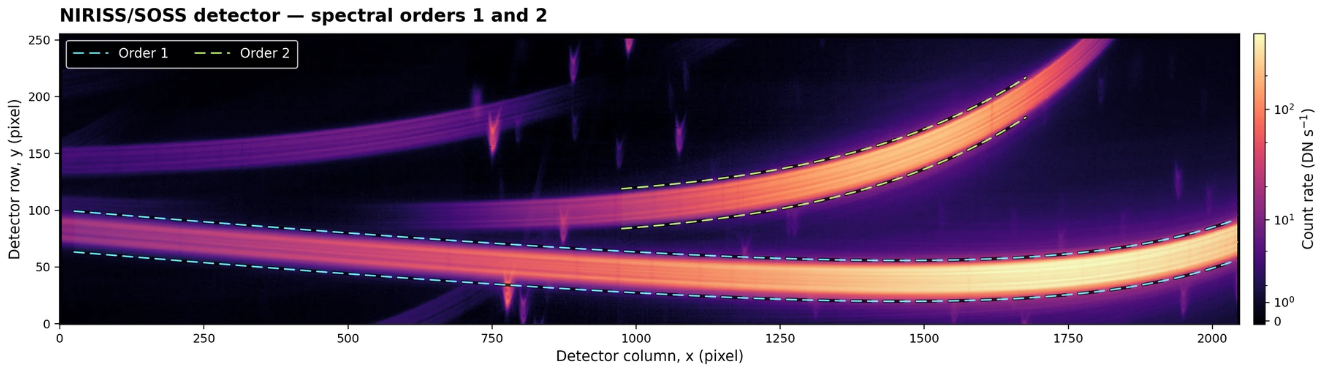

Figure 2: Median NIRISS/SOSS detector image showing the curved first- and second-order spectra. The cyan and green dashed boundaries indicate the detector footprints used for orders 1 and 2, respectively. An inverse hyperbolic sine (asinh) intensity stretch preserves the visibility of both the bright trace cores and their fainter wings. The colour bar reports the original count rate in DN s^{-1}.

time series of two-dimensional count-rate images [Albert et al., 2023].

The conventional analyses considered here follow an extraction-first workflow, summarised in Figure 3. The official JWST calibration pipeline or custom variants identify unusable data, correct saturation, bias, reference-pixel and nonlinearity effects, detect jumps, and fit the ramps. Subsequent processing assigns a wavelength and trace position to each pixel and treats flat-field response, low-frequency readout structure, background, field-star contamination, and bad pixels [Bell et al., 2022, Louie et al., 2025, Radica et al., 2023]. The precise order of these operations differs between pipelines. Background is sometimes estimated as part of extraction rather than only subtracted beforehand. In particular, JExoRES fits a residual background level simultaneously with the fluxes of the overlapping orders during its multi-order extraction [Holmberg and Madhusudhan, 2023]. This couples background estimation to spectral extraction, but not to the later time-dependent transit fit. A box extraction, optimal extraction, or forward extraction such as ATOCA then converts the two-dimensional images into one-dimensional spectra for each integration [Darveau-Bernier et al., 2022, Horne, 1986].

The extracted spectra are divided into wavelength channels, producing one light curve per channel. Each light curve is fitted with a transit model, but the treatment of residual time dependence is pipeline-specific rather than a universal step. In the published WASP-17 b comparison, the Ahsoka branch used Eureka! to fit polynomial time trends and a multiplicative noise term. The `transitspectroscopy/juliet` branch used a Matérn-3/2 Gaussian process, an offset, and an additional white-noise term. The Matérn process is a flexible stochastic model in which measurements close together in time are allowed to have correlated residuals. The supreme-SPOON branch used no additional time-systematics function for those light curves, but included an additive uncertainty term [Louie et al., 2025]. Finally, the fitted planet-to-star radius ratio in each channel is converted to transit depth and assembled as a function of wavelength to form the transmission spectrum.

Each stage contains choices that can alter the final spectrum. Examples include jump-detection thresholds, partially saturated ramps, bad-pixel masks, the order and location of the $1/f$ correction, scaling of the zodiacal-background model, treatment of dispersed field stars, trace profiles, extraction widths, wavelength bins, limb-darkening priors, and time-domain systematics models [Holmberg and Madhusudhan, 2023, Louie et al., 2025, Radica et al., 2023]. **The pipelines introduced in Section 1.2 implement different subsets of this chain:** Eureka! provides a modular framework from uncalibrated data through light-curve fitting [Bell et al., 2022]. Supreme-SPOON, now developed as exoTEDRF , **supreme-SPOON/exoTEDRF** includes SOSS-specific detector processing and extraction [Radica, 2024, Radica et al., 2023]. **,** `transitspectroscopy` provides routines for spectral tracing, extraction, and light-curve construction ; the WASP-17 b analysis paired it **paired** with `juliet` [Espinoza, 2022, Espinoza et al., 2019, Louie et al., 2025] . **for fitting [Espinoza, 2022, Espinoza et al., 2019], and** Ahsoka combines components of the official `jwst` calibration pipeline, supreme-SPOON, nirHiss, and Eureka! , while its WASP-17 b

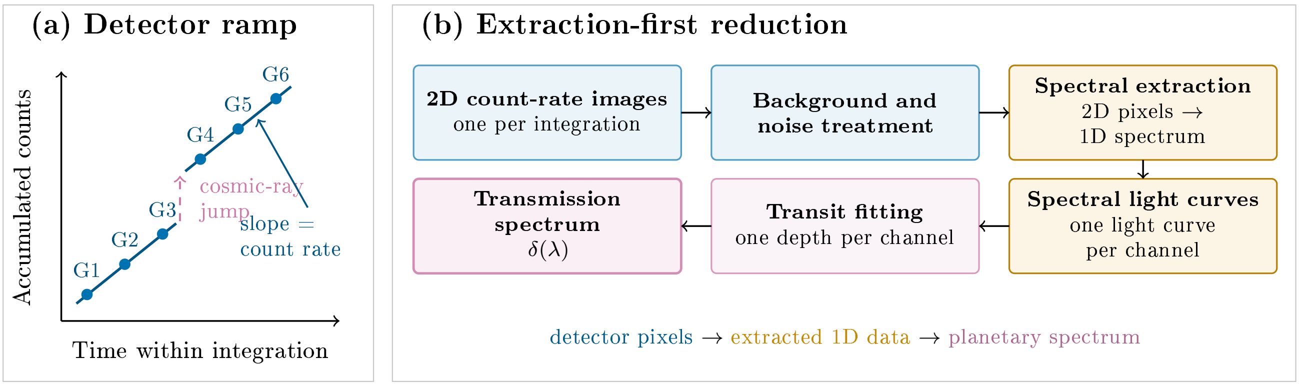

Figure 3: Shared conceptual backbone of extraction-first JWST/NIRISS SOSS reductions. (a) Nondestructive groups sample accumulated charge within an integration. A cosmic ray causes a discontinuity between groups, while the fitted slopes of the unaffected ramp segments estimate the pixel count rate. (b) Each integration produces a two-dimensional count-rate image. After background and noise treatment, spectral extraction compresses the detector pixels to a one-dimensional spectrum. Wavelength binning produces one light curve per channel, and transit fitting yields the wavelength-dependent depths that form the transmission spectrum. The exact implementation and ordering, including whether an operation occurs at group or count-rate level, differ between pipelines.

analysis uses several atmospheric forward models after reduction [Louie et al., 2025]. These tools overlap, but they are not interchangeable and do not necessarily implement the same stages.

SOSS also illustrates why extraction is a modelling problem. The first and second spectral orders overlap on the detector, so some pixels contain flux from both orders. ATOCA addresses this by constructing a linear model for every detector pixel and solving for the incident spectrum of both orders simultaneously [Darveau-Bernier et al., 2022]. Its accuracy depends on the trace profiles, wavelength maps, throughputs, and spectral-resolution kernels. APPLESOSS [Radica et al., 2022] estimates the spatial profiles needed by ATOCA directly from the observation. These algorithms are already forward models at the extraction stage, but they infer a stellar spectrum before the time-dependent transit inference performed later.

Every stage of this chain therefore admits defensible alternatives, and different combinations are in active use. The question is what the resulting spectra can and cannot establish. When independent reductions disagree, the result is scientifically useful but incomplete. It identifies sensitivity to methodological choices and may reveal an unmodelled uncertainty. For WASP-39 b NIRSpec data, different reductions shifted retrieved abundances by approximately one order of magnitude and altered the detectability of weaker species [Constantinou et al., 2023]. For the sub-Earth GJ 341 b, three reductions all supported the broad conclusion that no atmosphere was detected, yet two pipelines produced weak candidate features at different wavelengths and consequently favoured different molecular explanations [Kirk et al., 2024]. Such comparisons diagnose robustness, but the difference alone does not identify the more accurate spectrum.

Agreement answers a different question. The three WASP-17 b reductions agree closely over most of the SOSS bandpass [Louie et al., 2025], which is strong evidence that the result is reproducible across those implementations. Nevertheless, the analyses begin from the same observation, use common calibration reference files, and deliberately standardise several choices. Some also share software components [Bell et al., 2022, Louie et al., 2025, Radica et al., 2023]. Agreement therefore bounds the effect of the choices that differ, but it cannot exclude a bias caused by assumptions that are common to every analysis.

The limiting issue is the absence of ground truth. A real planet does not provide an independently known transmission spectrum against which a recovered spectrum can be scored. Agreement can establish reproducibility, and disagreement can expose sensitivity, but neither directly measures absolute recovery error. Model residuals are also insufficient because an incorrect decomposition of source, background, and instrument terms can still fit the observed pixels well. Absolute validation requires data for which the generating spectrum is known.

1.4 Structural limitations and the case for detector-level inference

Two structural choices in the conventional chain motivate this project. The first is background treatment. Published methods do not all treat the background identically. In the Ahsoka and supreme-SPOON implementations used for WASP-17 b, scaled versions of the STScI zodiacal-background model were subtracted before the extracted light curves were fitted. The `transitspectroscopy/juliet` branch followed the same general ordering [Louie et al., 2025]. JExoRES is an important exception: it refines a background component simultaneously with its multi-order spectral extraction [Holmberg and Madhusudhan, 2023]. It therefore models, rather than simply fixes, background at the extraction stage.

The central distinction is not whether a pipeline contains any background model. It is whether background and the wavelength-dependent transit are inferred together. In the published implementations considered here, background is either subtracted before extraction or refined jointly with extraction, but it is not inferred jointly with the transit signal in the same detector-pixel likelihood [Holmberg and Madhusudhan, 2023, Louie et al., 2025]. Once a one-dimensional spectrum has been produced, the later transit fit can account for background uncertainty only to the extent that its estimation error and covariance were carried into that product. The transit shape also cannot help revise how the earlier stage allocated detector counts between stellar source and additive background.

This distinction matters because background is additive while a transit is multiplicative on the stellar contribution. If a constant amount of flux is assigned incorrectly to the background, the inferred out-of-transit stellar flux changes and so does the fractional depth. If that error varies across the detector, it can alter spectral offsets, slopes, or local features. Classical two-dimensional spectroscopy already shows the value of modelling background in the native detector geometry before resampling [Kelson, 2003]. For slitless SOSS data, the problem also includes structured zodiacal light, field-star spectra, masked pixels, and $1/f$ readout structure [Holmberg and Madhusudhan, 2023, Radica et al., 2023].

The second choice is compression from detector pixels to one-dimensional spectra. A simple sum can lose the information needed to distinguish stellar light from background, defective pixels, or overlapping wavelengths. Extraction is not inevitably lossy, however. If the spatial profile, background, detector response, bad-pixel mask, and noise model are correct, the optimal-extraction formalism of Horne [1986] combines pixels with inverse-variance weights and can provide an efficient summary of the two-dimensional data. The difficulty arises when uncertain detector components are fixed during this compression, or when their covariance and response description are not carried forward. A later light-curve fit can then no longer revisit the original source-background or pixel assignment.

Spectro-perfectionism, introduced by Bolton and Schlegel [2010], makes this condition explicit. It models the detector image with the full two-dimensional point-spread function and shows that a one-dimensional product can preserve the pixel-level likelihood if the extracted spectrum is delivered with an appropriate covariance and resolution description [Bolton et al., 2012]. ATOCA applies a related pixel model to overlapping SOSS orders [Darveau-Bernier et al., 2022]. Conversely, if the trace, wavelength mixing, background, bad-pixel treatment, or order decomposition is fixed incorrectly, and the resulting covariance is not carried forward, the later light-curve fit cannot recover the discarded information.

Even a forward-modelled extraction normally separates the stellar-spectrum estimate from the later transit fit. ATOCA treats the incident spectrum as the unknown quantity at each integration, while time-dependent transit parameters are inferred after extraction [Bell et al., 2022, Darveau-Bernier et al., 2022, Radica et al., 2023]. This ordering is practical and may be entirely adequate. It does, however, prevent the transit model from helping to separate stellar light, additive background, overlapping orders, and detector response within a single likelihood. Whether that separation causes a meaningful bias for SOSS transmission spectra is an empirical question, not an assumption of this project.

The validation problem motivates a detector-level benchmark. A forward injector generates time-series observations with a known transmission spectrum, stellar signal, instrument response, additive background, noise, and detector flags. Every recovery method receives the same synthetic detector cube and the same declared auxiliary information. The injected spectrum remains unavailable to the recovery process until its prediction has been fixed. Recovery error can then be measured directly as a function of wavelength, spectral morphology, order, signal level, background structure, and detector condition.

NOVA is the complementary recovery method. Instead of extracting a stellar spectrum and fitting spectral light curves in separate stages, NOVA evaluates a forward model in detector space. The wavelength-dependent transmission spectrum is shared across the useful SOSS orders, propagated through the instrument response, and fitted jointly with additive background and nuisance components. Only stellar light is multiplied by the transit model. This preserves the physical distinction between source and background and allows information from multiple detector pixels and orders to constrain the same transmission spectrum.

The benchmark and NOVA answer different parts of the problem. NOVA cannot validate itself, because a successful fit to pixels does not prove that the stellar, background, and transit components were separated correctly. The benchmark supplies the missing truth and permits a controlled comparison with extraction-based methods. It can also reveal conditions under which established pipelines are already unbiased, conditions under which NOVA improves recovery, and conditions under which every tested method fails because the instrument or injector model is inadequate.

1.5 Research question and objectives

This project asks:

> *How accurately and robustly can JWST/NIRISS SOSS transmission spectra be recovered from detector-level time-series observations, and can joint detector-level inference reduce biases associated with background treatment and extraction-first analysis?*

The research question is divided into four objectives:

1. develop an independent detector-level benchmark with known injected transmission spectra;
2. develop NOVA as a joint detector-level inference method for NIRISS/SOSS transmission spectroscopy;
3. compare NOVA with extraction-based analyses under controlled observing conditions and atmospheric scenarios; and
4. identify the detector, calibration, and modelling assumptions that limit recovery accuracy and determine how those limitations propagate into atmospheric interpretation.

The present assessment defines the problem, documents the current method, and sets out the experiments required to answer the research question. WASP-17 b is the first development case. It will be followed by controlled variations in background, spectral morphology, noise, masking, instrument response, and target properties, followed by application to additional observations. Results are reported only where the associated injector and recovery checks have been completed.

2 Methods and Benchmark Design

2.1 Experimental design and scope

The project uses a controlled recovery experiment. A detector-level injector generates a synthetic SOSS time series with a known transmission spectrum. NOVA and one or more extraction-based analyses receive the same detector cube, time grid, uncertainty array, data-quality information, and declared calibration products. Each method produces a fixed spectral prediction before the injected spectrum is made available for scoring. The experiment therefore separates model development from absolute evaluation.

The initial geometry is based on the JWST/NIRISS SOSS transit of WASP-17 b. This provides realistic trace positions, wavelength coverage, time sampling, detector uncertainties, masks, background structure, and overlapping-order geometry. The atmospheric spectrum is synthetic and is not inferred from the real observation. The first benchmark contains a zero-feature control and structured spectra with broad and localised features. Later experiments will vary source brightness, background amplitude and morphology, detector defects, spectral morphology, and response mismatch. This staged design isolates failure modes before combining them in a larger factorial benchmark.

The unit of the NOVA likelihood is one retained physical detector pixel in one integration. The union of the useful order masks defines the pixel set. If more than one order contributes to the same pixel, those predicted contributions are summed before the single observed value is evaluated. Non-finite measurements, non-positive uncertainties, and pixels carrying the JWST `DO_NOT_USE` flag are excluded by a fixed mask.

2.2 Ground-truth detector-level injector

The injector begins with an out-of-transit stellar source and a separately estimated additive background. A known wavelength-dependent transit is applied only to the stellar component. The attenuated source is projected through a generator response for each order and the order contributions are added in overlap pixels. The background, noise, masks, and required detector metadata are then combined with the source to produce a calibrated detector-rate cube.

The source and background are estimated jointly from out-of-transit integrations using separate spatial and temporal components. The currently selected source solution uses a shared stellar continuum and a four-mode additive background model. Temporal fold separation is used when estimating source wings and response components: a component fitted on one subset is tested on the held-out subset before it can enter the injector. This is intended to prevent the injector from reproducing arbitrary in-sample structure.

Injector and recovery products are separated by construction. Recovery code can read only the delivered cube and the preregistered auxiliary products. It cannot read the atmospheric spectrum, generator-only response products, or scoring code. Hashes identify the delivered arrays, configuration, masks, and response products. The injected spectrum is loaded only after the prediction and its provenance record have been closed. This procedure belongs to the benchmark construction because it defines what counts as a valid ground-truth experiment.

2.3 The NOVA detector-level forward model

NOVA models the calibrated detector time series directly. For integration t and physical detector pixel p, the current response-aware mean model is

$$\widehat{y}_{tp} = B_{tp}(\boldsymbol{\alpha}) + \sum_{o} \left[\boldsymbol{R}_o^{\mathrm{rec}} \left\{\boldsymbol{S}_t(\boldsymbol{\beta}) \odot \boldsymbol{T}_{o,t}(\boldsymbol{\theta})\right\}\right]_p, \quad (2)$$

where B_{tp} is additive background, $\boldsymbol{R}_o^{\mathrm{rec}}$ maps the fine wavelength grid of order o to detector pixels, $\boldsymbol{S}_t$ is the shared stellar source, and $\boldsymbol{T}_{o,t}$ is the exposure-integrated transit. The symbol $\odot$ denotes element-wise multiplication on the wavelength grid. The instrument response includes trace geometry, throughput, wavelength mapping, line-spread information, and the mixing of wavelengths within each pixel.

The transmission spectrum is represented by $K = 160$ shared coefficients,

$$D(\lambda) = \sum_{j=1}^{K} B_j(\lambda) D_j, \quad (3)$$

with positive depths enforced by a smooth bounded coordinate transformation. Both orders evaluate the same physical spectrum on their own wavelength grids. Any order-discrepancy term is restricted to a diagnostic branch and is not part of the primary shared-spectrum model. This allows overlapping orders to constrain the same atmospheric quantity without treating repeated detector measurements as independent copies.

For depth D in parts per million, the radius ratio is $k = \sqrt{D \times 10^{-6}}$. NOVA evaluates an exact Keplerian transit with wavelength-dependent quadratic limb darkening [Mandel and Agol, 2002]. Each detector integration is modelled as a Gauss-Legendre average across its finite exposure rather than at a single mid-exposure time. A convergence test selected 21 quadrature points, for which the maximum difference relative to a 61-point reference was below 0.01 ppm across the tested depths and limb-darkening locations.

2.4 Background and nuisance model

The stellar continuum and background coefficients are linear conditional on the nonlinear transit state. The current response-aware stellar model is shared in physical wavelength space and is propagated separately through each order response. Earlier development branches tested order-tagged and column-local continua; those branches remain useful controls but are not the current primary model.

The additive background uses eight spatial modes crossed with two temporal modes, giving 16 fitted coefficients. Off-trace pixels provide an external anchor on these modes. The background is never multiplied by the transit. Consequently, a change in transit depth cannot be compensated by attenuating the background in the same way as the star. Source and background are nevertheless partly degenerate, so the fit records nuisance rank, singular values, conditioning, source positivity, and held-out background diagnostics.

The current likelihood does not include a generic Gaussian-process model for time-correlated residuals. The fitted background modes describe additive detector structure, while the Huber weights introduced below reduce the influence of outlying residuals. Neither is equivalent to the Matérn process used in some extracted light-curve analyses. Residual temporal correlation is therefore retained as a diagnostic and as a possible future extension of the likelihood.

Quadratic limb darkening is parameterised using physically bounded coordinates [Kipping, 2013],

$$u_1 = 2\sqrt{q_1}q_2, \qquad u_2 = \sqrt{q_1}(1 - 2q_2). \tag{4}$$

Smooth wavelength-dependent deviations are represented by six knots per order around stellar-atmosphere predictions. Separate residual blocks constrain the deviation from theory and excessive wavelength curvature. These constraints are reported separately from the detector likelihood.

2.5 Profiled optimisation and robust fitting

Let $\boldsymbol{\theta}$ contain the nonlinear depth and limb-darkening parameters and $\boldsymbol{\eta}$ the linear stellar and background coefficients. At fixed $\boldsymbol{\theta}$, NOVA solves

$$\widehat{\boldsymbol{\eta}}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\eta}} \frac{1}{2} \|\boldsymbol{W}[\boldsymbol{y} - \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{\eta}]\|_2^2 + \frac{1}{2} \|\boldsymbol{L}\boldsymbol{\eta}\|_2^2. \tag{5}$$

This variable-projection step [Golub and Pereyra, 1973] removes the linear variables from the nonlinear search while retaining their exact response to $\boldsymbol{\theta}$. The profiled Jacobian includes movement of the fitted stellar and background coefficients. The production implementation streams detector rows into sparse factors and a small background Schur complement, avoiding a visit-sized dense Jacobian.

The nonlinear profiled problem is solved with bounded trust-region reflective least squares [Branch et al., 1999]. Robustness is provided by Huber iteratively reweighted least squares [Huber, 1964] with threshold 1.345. Each robust cycle recomputes detector weights from standardised residuals, refits the model, and stores an immutable prediction and diagnostic record. The standard policy permits 25 cycles before a soft limit and 50 before a hard stop. A final fit using the recomputed terminal weights is required before a start can be accepted.

Two deterministic starts are fitted independently. Convergence is not assigned from an optimiser return flag alone. It requires stable nuisance rank, valid physics, successful bounded optimisation, small weight and objective changes, small spectral changes, first-order criteria, and agreement between starts. Current multistart tolerances are below 0.1 ppm on the common-40 spectrum, below 0.5 ppm on the F2 report grid, and below 10^{-8} in relative robust objective. Failure of any required gate is retained and reported.

2.6 Extraction-based comparison analyses

The first external comparison uses an authentic Ahsoka reduction of the same delivered cube. It follows the analysis topology used for the published WASP-17 b spectrum, including detector preparation, spectral extraction, spectroscopic light curves, and channel-level transit inference [Louie et al., 2025]. The diagnostic comparison uses 200 walkers, 1,000 steps, and a 500-step burn-in, with the squared posterior-median radius ratio reported as transit depth.

Fairness is defined at delivery rather than by forcing the two methods to use the same internal representation. The pixel values, uncertainties, flags, time grid, and declared wavelength support are identical. Unsupported bins are excluded symmetrically. Each method retains its own intended nuisance model and inference strategy. Later benchmark rounds will include Eureka!, exoTEDRF, transitspectroscopy, and the official JWST pipeline with ATOCA extraction [Bell et al., 2022, Darveau-Bernier et al., 2022, Louie et al., 2025, Radica, 2024]. Results will be reported both for each normal end-to-end analysis and with a common downstream light-curve fitter. The first comparison measures the spectra that a user would obtain in practice; the second isolates differences introduced before the light-curve model. Controlled variants of background and extraction choices will distinguish a NOVA-specific comparison from a broader claim about detector-level and extraction-first analysis.

2.7 Evaluation metrics

Let $\widehat{D}_i$ and D_i^{true} be the recovered and injected depths on a common reporting grid and $e_i = \widehat{D}_i - D_i^{\text{true}}$. The absolute recovery error is

$$\mathrm{RMSE}_{\text{abs}} = \left(\frac{1}{N} \sum_i e_i^2 \right)^{1/2}. \tag{6}$$

The morphology error removes the mean residual,

$$\mathrm{RMSE}_{\text{morph}} = \left[\frac{1}{N} \sum_i (e_i - \bar{e})^2 \right]^{1/2}, \tag{7}$$

so that a constant offset can be distinguished from incorrect spectral shape. Mean and median bias, feature-amplitude error, order-overlap consistency, and order-specific residuals are also reported. The primary comparison grid is a common $R = 100$ grid, with common-40 and F2 grids retained for regression and diagnostic reporting.

Uncertainty calibration will be evaluated through standardised residuals, empirical coverage of injected depths, and repeated noise realisations. Detector residuals, robust weights, rank, conditioning, boundary activity, and convergence diagnostics are evaluated alongside the spectral scores. A low RMSE is not sufficient if a fit fails its preregistered numerical or physical validity gates.

3 Results and Current Status

This section reports the state of the project frozen on 21 August 2026. The results are development and screening evidence. None is presented as a final measurement of WASP-17 b, and the current benchmark remains `science_ready=0`.

3.1 Injector validation

The supported-wing detector injector passed its internal identity and held-out checks. The maximum difference between the constructed comparison cubes at the pixel level was 5.945×10^{-14} DN s^{-1}, and the maximum identity difference in the order-overlap region was 5.232×10^{-14} DN s^{-1}. No background-only pixels changed. Increasing the exposure-integration rule from 41 to 61 points altered the delivered spectral prediction by 0.013116 ppm. These tests establish numerical identity of delivery and adequate exposure quadrature for the diagnostic case.

The supported stellar-wing correction was estimated using temporal fold A and tested on fold B. The held-out residual wing fraction was 1.6383×10^{-4}, while the source-fraction difference between the two folds was 1.02119×10^{-3}. This supports the use of the correction within the declared detector support. A genuine Order-1 response hole remained unsupported. Three central bins and two preregistered partial edge bins around that region were therefore excluded from both recovery methods rather than reconstructed with unvalidated information.

Table 1: Selected closed development results on the common-$R = 100$ grid. These are diagnostic scores, not final scientific results.

Branch	Abs. RMSE (ppm)	Morph. RMSE (ppm)
NOVA V28-R2	203.631	174.655
NOVA V30-R8	211.709	189.095

3.2 Transmission-spectrum recovery

Successive response-aware NOVA branches showed that instrument response and wavelength support dominate the current error budget. The best completed result in the directly compared V26 to V30 development set was the physically contracted K155 support branch, with 203.631 ppm absolute and 174.655 ppm morphology RMSE on the common-$R = 100$ grid. A later Fold-B response branch completed at 211.709 ppm absolute and 189.095 ppm morphology RMSE. Its two deterministic starts agreed to well below 1 ppm, showing that reproducibility of the numerical endpoint did not guarantee a more accurate physical solution.

The residual in the Fold-B branch was dominated by Order 1. Two bins on the flanks of the 1.03 to 1.10 μm support gap contributed approximately 107.5 ppm in quadrature. Removing only those bins left about 183.7 ppm absolute RMSE. Order 2 below 0.84 μm recovered more accurately than Order 1 despite the higher nominal signal-to-noise ratio of Order 1. This localised the next methodological problem to response, mask, support, wavelength, or source allocation rather than to a uniform lack of signal.

3.3 Cross-method comparison

An end-to-end diagnostic comparison was completed on one opened structured atmospheric case using the supported-wing injector. NOVA obtained an absolute RMSE of 281.873 ppm and morphology RMSE of 255.725 ppm on the common-$R = 100$ grid. The authentic Ahsoka analysis obtained 756.498 ppm and 352.440 ppm, respectively. NOVA's mean bias was +84.301 ppm and Ahsoka's was −450.844 ppm. Feature-amplitude errors differed in sign and wavelength, showing that the comparison cannot be reduced to a single constant offset.

The lower NOVA error in this case is encouraging but does not establish general superiority. The NOVA reducer classified the two-start solu-

Table 2: Opened one-case diagnostic comparison on the common-$R = 100$ grid.

Method	Abs. RMSE (ppm)	Morph. RMSE (ppm)	Bias (ppm)
NOVA	281.873	255.725	+84.301
Ahsoka	756.498	352.440	−450.844

tion as non-strict, so it is excluded from a headline scientific comparison. The source and background decomposition was also specific to this injector, and only one atmospheric case was opened. The result demonstrates that the benchmark can discriminate methods and diagnose failure structure. It does not yet answer how either method performs across targets, noise realisations, backgrounds, or response mismatch.

3.4 Instrument-response and support limitations

The response-aware development changed the interpretation of earlier results. The first injector and recovery shared too much spatial-response structure, so their strong closure was not an independent end-to-end test. Later external response models removed this matched structure and exposed supported stellar wings, source-background allocation, and the Order-1 gap as limiting factors. An out-of-transit-median spatial-profile substitution also risked allocating background morphology to the star and was rejected from the selected branch.

The current response model replaces legacy SPECTRACE calibration, rebuilds the generator mask consistently, and constructs Fold-A, Fold-B, and full-data response variants. The associated factorial experiment is designed to separate calibration and support changes from temporal-fold choice. At the handoff freeze, these responses and validation products had been constructed, but the final fair injector regeneration and downstream recovery were not complete.

3.5 What has and has not been established

The work has established four points. First, a detector-level SOSS cube can be delivered identically to NOVA and an authentic extraction-based analysis. Second, the injected spectrum can be withheld until a prediction is fixed and then used to compute absolute and morphology errors. Third, NOVA can jointly fit a shared transmission spectrum and additive background at detector level. Fourth, response support and source-background allocation can dominate the recovered spectrum even when the optimisation is reproducible.

Several claims remain open. The current work has not demonstrated strict NOVA convergence on the fair response-consistent injector, calibrated frequentist or posterior uncertainties, superiority across a preregistered ensemble, or an improved atmospheric inference on real data. It has also not completed an independent limb-darkening cross-check or a multi-pipeline benchmark. These open claims define the future research plan rather than weaknesses to be hidden by the current diagnostic scores.

4 Research Plan and Future Timeline

The remainder of the PhD will turn the current development system into a tested scientific method, determine where reduction choices change atmospheric interpretation, and then combine validated observations across visits, instruments, and observing geometries. The programme is organised into five work packages. The timeline begins with the remaining part of 2026 Q3 and contains future work only.

4.1 WP1: Strict NOVA and injector closure

The first priority, to be completed by 30 September 2026, is to finish the response-consistent injector, obtain strict multistart convergence for NOVA, and close the initial WASP-17 b benchmark. The stellar continuum will be refitted with the same generator response used to construct each synthetic cube. The cube will then be delivered unchanged to the frozen recovery methods. Response support, masks, time sampling, and auxiliary information will be checked before any recovery is scored.

The benchmark will be expanded from the current opened case to a preregistered matrix. At minimum, it will include a featureless spectrum, broad molecular structure, localised features, several signal amplitudes, multiple noise realisations, background amplitudes and morphologies, and controlled response mismatch. Injector-only checks will test pixel identity, flux accounting, fold-held-out prediction, exposure integration, and mask symmetry. Recovery success will require strict convergence, agreement between deterministic starts, valid nuisance rank and physics, and fixed-prediction scoring.

The deliverable is a versioned benchmark package with documented inputs, comparison grids, metrics, and provenance. Its primary scientific result will be a map of recovery error as a function of injected and detector condition, not a single favourable spectrum.

4.2 WP2: Robustness, pipeline benchmark, and methods paper

NOVA will next be run on out-of-transit data from additional visits. This tests whether the inferred spatial response and additive background model are robust to changes in pointing, background morphology, detector state, and illumination history. Synthetic small problems will continue to test the profiled Jacobian against a dense reference, while visit-scale runs will test numerical stability, checkpoint reproducibility, and computational scaling.

The expanded comparison will include Ahsoka, Eureka!, exoTEDRF, formerly supreme-SPOON, transitspectroscopy, and the official JWST calibration pipeline with ATOCA extraction [Bell et al., 2022, Darveau-Bernier et al., 2022, Louie et al., 2025, Radica, 2024]. The official pipeline is a calibration and extraction baseline rather than a complete atmospheric analysis, so its extracted time series will be passed to a declared light-curve model. Two comparison tracks will be reported. The first will preserve each pipeline's normal end-to-end analysis. The second will pass the extracted products to a common light-curve fitter to separate detector and extraction effects from differences in transit modelling.

Ablation experiments will fix or jointly fit the background, replace the response-aware source with simpler profiles, remove robust weighting, alter the spectral basis resolution, fit orders separately, and reproduce selected extraction-first compressions. Each ablation will use the same injected cube and scoring protocol. This will distinguish the effect of joint detector modelling from the effect of a particular nuisance basis or optimiser.

The principal deliverable from WP1 and WP2 is the NOVA and benchmark methods paper, targeted for submission in 2027 Q1. The associated software release will contain the minimum reproducible injector, recovery configurations, closed predictions, and score tables required to reproduce the paper figures.

4.3 WP3: Multi-target NIRISS/SOSS validation

After the method passes the synthetic benchmark, it will be applied to additional public NIRISS/SOSS transmission observations. Targets will be selected to span stellar brightness, background conditions, transit depth, atmospheric signal amplitude, and known reduction challenges. Priority will be given to observations with at least two published reductions or publicly reproducible pipelines because they provide an external reproducibility baseline.

For real observations, the unknown planetary spectrum prevents an absolute score. Validation will therefore combine detector-domain predictive checks, order-overlap consistency, independent reductions, injection into observation residuals, and sensitivity to the source and background bases. The scientific question will be whether NOVA changes repeatable atmospheric features or mainly changes offsets and uncertainties. Atmospheric retrievals will be performed only after the spectral recovery is fixed, so that detector-model uncertainty can be traced into abundance and cloud constraints.

The SOSS analysis will be frozen by 30 June 2027. A fixed number of papers is not promised in advance because the publication decision is itself scientific. A target-specific paper will be prepared only when the recovered spectrum changes repeatably and the change materially alters an atmospheric inference. This avoids dividing null results into artificial papers while allowing several focused papers if several planets show consequential differences. The analysis freeze, rather than journal acceptance, is the controllable mid-2027 milestone; target-paper submission can continue into 2027 Q3 while the next work package begins.

4.4 WP4: Emission, phase curves, and other instruments

After the SOSS transmission programme, the temporal model will be extended first to secondary-eclipse emission and then to phase curves. Remaining initially with NIRISS/SOSS isolates the change in observing geometry from the change in instrument. The next step will adapt the validated strategy to NIRSpec and MIRI, with a benchmark specific to each detector and observing mode rather than an assumption that a SOSS response model transfers directly.

This extension is supported by existing same-planet observations. WASP-17 b has separate

NIRISS/SOSS transmission and eclipse measurements, together with consecutive NIRSpec transit and eclipse observations [Gressier et al., 2024, Louie et al., 2025, Lustig-Yaeger et al., 2025]. WASP-121 b has published NIRSpec and NIRISS phase curves as well as a NIRSpec transmission analysis [Gapp et al., 2025, Mikal-Evans et al., 2023, Splinter et al., 2025]. GJ 1214 b provides a contrasting sub-Neptune case with a MIRI phase curve and a panchromatic HST, NIRSpec, and MIRI transmission analysis [Kempton et al., 2023, Ohno et al., 2025]. A systematic literature and archive review will distinguish repeated visits in one mode, different instruments measuring the same geometry, and genuinely complementary transmission, eclipse, and phase-resolved emission observations.

The final scientific objective is a joint analysis of selected planets across instruments and geometries. The goal is not simply to concatenate published spectra. Reduction uncertainty, visit variability, shared planetary parameters, and geometry-specific atmospheric weighting must be represented explicitly so that the joint retrieval can test whether the combined data tighten atmospheric constraints without hiding inconsistencies.

4.5 WP5: Synthesis and thesis writing

The thesis will integrate the literature review, benchmark design, NOVA method, controlled comparisons, multi-target SOSS results, cross-instrument extensions, and joint atmospheric analysis. Writing will proceed alongside the papers so that methods, assumptions, negative results, and provenance are captured while the analyses are active. The 3.5-year research phase ends on approximately 27 April 2029 after accounting for the one-month interruption. The formal writing-up period then runs to 28 October 2029 and is reserved for thesis completion, submission, corrections, and viva preparation rather than new method development.

4.6 Timeline, milestones, and dependencies

The dependency structure is shown in Figure 4. The response-consistent injector and strict NOVA convergence precede the headline comparison. Multi-visit testing and the expanded pipeline benchmark can proceed in parallel once the recovery interfaces are frozen. Target selection can begin while the methods paper is written, but atmospheric claims are made only after the benchmark gates are passed. Emission and phase-curve work begins after the SOSS analysis freeze, while the literature and archive audit can start in parallel. Instrument-specific extensions precede the joint retrieval.

Progress will be reviewed quarterly against closed deliverables rather than software activity. A milestone is complete only when its data products, configuration, diagnostics, and written interpretation are archived together.

4.7 Risks and contingency plan

The main technical risk is that the instrument response remains insufficiently accurate for strict end-to-end recovery. This will be addressed with held-out response validation, wavelength-local diagnostics, independent calibration routes, and simulations in which the response is exactly known. If real-response uncertainty remains dominant, the thesis can still quantify the required response accuracy and show how it propagates into the recovered spectrum.

The second risk is computational cost. Variable projection, streamed derivatives, checkpointed robust cycles, and staged benchmark grids reduce this risk. Small diagnostic cases will be required before full visit-scale runs. The third risk is that NOVA does not outperform extraction-based analyses. That outcome remains scientifically useful if tested fairly: the project would establish the conditions under which extraction is sufficient and place an empirical upper bound on the benefit of joint detector inference.

The final scope risk is overextension. Cross-instrument and cross-geometry inference is the central long-term objective, but its breadth is adjustable. If time is constrained, the synthesis can focus on one well-observed planet and two complementary instruments rather than an incomplete survey of many systems. The scientific requirement is a defensible joint inference with propagated reduction uncertainty, not a predetermined number of targets or papers.

5 Conclusions

This project addresses a validation problem created by high-precision JWST transmission spectroscopy. Different reductions can reveal sensitivity to analysis choices, and agreement can establish reproducibility, but real observations do not

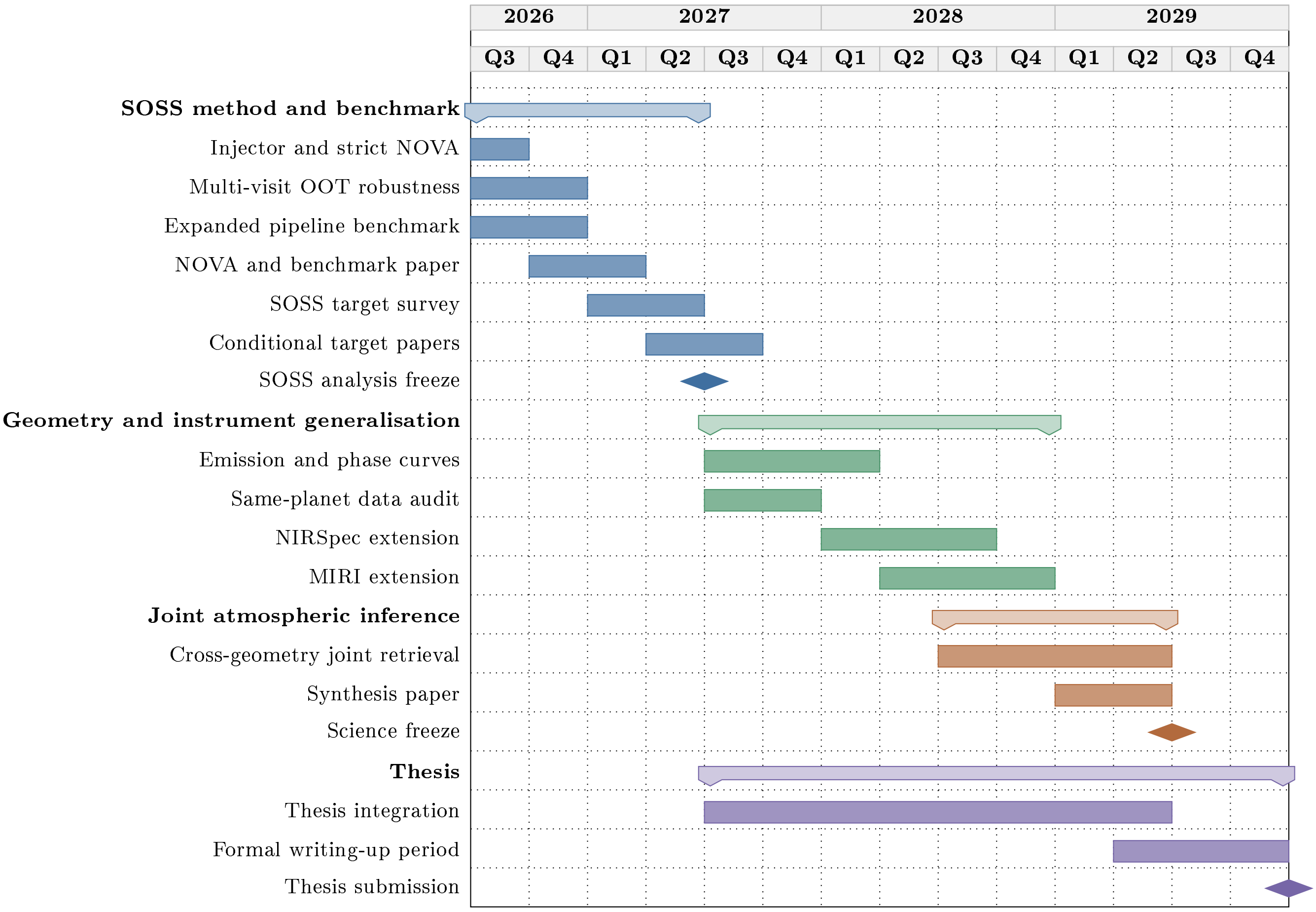

Figure 4: Future-only PhD plan in quarters. The first column covers the remaining part of 2026 Q3. The SOSS analysis is frozen by 30 June 2027; conditional target manuscripts may continue into Q3. The formal writing-up period runs from approximately 27 April to 28 October 2029.

provide the known spectrum required to measure absolute recovery error. The proposed solution combines a realistic detector-level injector with NOVA, a joint detector-level forward model for the stellar transit, overlapping SOSS orders, instrument response, and additive background.

At the present stage, the benchmark can deliver identical synthetic detector data to NOVA and an authentic extraction-based analysis, keep the injected spectrum outside the recovery, and score fixed predictions afterwards. The initial opened case shows that the methods can produce measurably different absolute and morphology errors. It also shows why strict numerical and physical gates are needed: the lower NOVA error was obtained by a fit that did not satisfy all convergence requirements. Response support and source-background allocation, rather than statistical noise alone, are the main current limitations.

The research question is therefore only partly answered. Detector-level recovery is feasible and has diagnostic value, but improved accuracy and robustness have not yet been demonstrated across a fair preregistered ensemble. The remaining PhD programme will complete that test, identify the conditions under which NOVA helps, apply the validated framework to additional SOSS observations, and propagate spectral-recovery differences into atmospheric interpretation. The subsequent programme will extend the validated strategy to emission, phase curves, and other JWST instruments, with the final aim of combining complementary observations of the same planets into more reliable atmospheric constraints.

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A Detailed NOVA Model and Numerical Implementation

A.1 Detector-level forward equations

Write the retained detector data as $\boldsymbol{y} \in \mathbb{R}^N$ with diagonal standardisation matrix $\boldsymbol{W}$. For nonlinear state $\boldsymbol{\theta}$, the model is linear in the stellar-continuum coefficients $\boldsymbol{\beta}$ and background coefficients $\boldsymbol{\alpha}$,

$$\widehat{\boldsymbol{y}}(\boldsymbol{\theta}, \boldsymbol{\eta}) = \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{\eta}, \qquad \boldsymbol{\eta} = \begin{bmatrix} \boldsymbol{\beta} \\ \boldsymbol{\alpha} \end{bmatrix}. \tag{8}$$

Each row corresponds to one physical pixel and integration. Contributions from all orders assigned to that pixel are summed in the row before comparison with the data. The response matrices remain defined on a fine wavelength grid, while the fitted stellar continuum and planetary depth use lower-dimensional physical bases. This separates instrument sampling from the number of fitted coefficients.

A.2 Transit and limb-darkening parameterisation

Positive atmospheric depths are represented in unconstrained optimiser coordinates and mapped smoothly into the allowed physical interval. Quadratic limb-darkening coefficients are obtained from bounded q_1 and q_2 coordinates using Equation 4. The transit engine evaluates a Keplerian orbit and integrates the model across every exposure. Unphysical depth, geometry, or limb darkening causes a diagnostic failure rather than a fallback model.

The primary atmospheric basis is shared between orders. A smooth two-coefficient order-discrepancy model exists only as a diagnostic for response mismatch. It is constrained to prevent the two orders from becoming independent spectra and is excluded from the primary result unless a preregistered comparison requires it.

A.3 Variable projection and additive background

For a fixed robust-weight cycle, the profiled residual is

$$\boldsymbol{r}(\boldsymbol{\theta}) = \boldsymbol{C}(\boldsymbol{\theta})\widehat{\boldsymbol{\eta}}(\boldsymbol{\theta}) - \boldsymbol{d}, \tag{9}$$

where $\boldsymbol{C}$ and $\boldsymbol{d}$ include detector standardisation and any declared linear regularisation rows. If $\boldsymbol{C}_a = \partial \boldsymbol{C}/\partial\theta_a$, movement of the profiled linear solution satisfies

$$(\boldsymbol{C}^\mathsf{T}\boldsymbol{C})\boldsymbol{\eta}_a = -\left(\boldsymbol{C}^\mathsf{T}\boldsymbol{C}_a\widehat{\boldsymbol{\eta}} + \boldsymbol{C}_a^\mathsf{T}\boldsymbol{r}\right), \tag{10}$$

and the exact profiled derivative is

$$\boldsymbol{r}_a = \boldsymbol{C}_a\widehat{\boldsymbol{\eta}} + \boldsymbol{C}\boldsymbol{\eta}_a. \tag{11}$$

This derivative includes the response of both stellar and background coefficients. Treating the fitted nuisance coefficients as fixed would omit part of the derivative and can alter the nonlinear step.

The production solve exploits the sparse stellar-response block and eliminates it into a 16-dimensional background Schur complement. Verification-sized problems are also solved with dense rank-revealing linear algebra. Agreement between the structured and dense predictions, objectives, and Jacobians is a release requirement.

A.4 Huber IRLS and numerical safeguards

For standardised detector residual z, NOVA uses the Huber loss [Huber, 1964] with threshold $\delta = 1.345$,

$$\rho_\delta(z) = \begin{cases} \frac{1}{2}z^2, & |z| \le \delta, \\ \delta(|z| - \frac{1}{2}\delta), & |z| > \delta. \end{cases} \tag{12}$$

IRLS converts this loss into fixed-weight profiled least-squares cycles. The accepted prediction must pass a final confirmation using newly recomputed weights. Every cycle records objective components, robust-weight change, spectral change, gradient measures, bounds, nuisance rank, conditioning, and prediction hashes.

The optimiser rejects invalid physical or coordinate proposals. Two independent starts must converge to compatible spectra and objectives. A deterministic material-descent audit tests feasible local directions after nominal termination. These safeguards were added because small objective changes and an optimiser success flag were found insufficient to establish a unique numerical endpoint.

B Benchmark and Reproducibility Protocol

B.1 Injector and recovery separation

The benchmark is divided into generator, delivery, recovery, and scoring stages. The generator owns the atmospheric truth and generator-only response. Delivery creates the detector cube and a manifest of the products visible to recovery. Recovery owns the model configuration, checkpoints, diagnostics, and fixed prediction. Scoring reads the closed prediction and truth only after recovery is complete. No score is used to choose an optimiser start, checkpoint, nuisance basis, or response branch.

B.2 Prediction closure and truth access

A valid prediction closure contains the spectrum on all declared reporting grids, detector-domain prediction identity, convergence status, configuration and source hashes, and an immutable timestamped manifest. Failed and non-strict fits are still scored for diagnosis but cannot enter the primary comparison. Any model change after truth access creates a new benchmark branch and requires a new hidden case.

B.3 Comparison metrics and reporting convention

Scores are reported in ppm on a common $R = 100$ grid and on retained regression grids. Unsupported wavelength bins are listed and excluded symmetrically. Every table distinguishes a constant bias from morphology error. Aggregate rankings will be reported across the preregistered ensemble, together with the distribution of case-level errors, so that one favourable case cannot determine the conclusion.

Table 3: Minimum report for each benchmark case.

Category	Required record
Delivery	Cube, uncertainties, data-quality flags, time grid, wavelength support, response and configuration identities
Numerical validity	Termination state, robust-weight closure, multistart agreement, rank, conditioning, bounds, and physical-validity gates
Spectral accuracy	Absolute and morphology RMSE, mean and median bias, feature errors, and order-specific residuals
Uncertainty	Standardised residuals, empirical interval coverage, repeated noise realisations, and wavelength covariance where available
Detector diagnostics	Residual maps by time, order, detector region, and background mode, including masked and unsupported regions
Scope statement	Primary, diagnostic, failed, or non-strict classification and a statement of claims that the case can and cannot support

B.4 Software and data provenance

Every benchmark release records content hashes for code, configuration, response, stellar source, background, detector arrays, masks, closed predictions, and score tables. Cluster-only arrays are represented by authenticated manifests when they cannot be included in the portable archive. Superseded and failed branches remain available with an explicit status because they document rejected assumptions and prevent accidental reuse as selected results.